

Mobility Lens Application Report

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Package: edu.bu.mobilitylens Emulator: Medium Phone (sdk_gphone64_x86_64)

SDK Configuration

- minSdk (24): The lowest Android version the app supports (Android 7.0). I selected 24 to ensure broad device compatibility (over 95% of active devices) while allowing the use of modern Jetpack Compose features without complex legacy workarounds.
- targetSdk (36): The API level the application was actively tested against. It signals the Android system to apply Android 15's specific runtime behaviors and visual standards to the app.
- compileSdk (36): The SDK version Gradle uses to compile the code. This determines which Android APIs are accessible in Kotlin during development.

Project Architecture

- MainActivity (app/src/main/java/edu/bu/mobilitylens/MainActivity.kt): The application's entry point. It initializes the window and hosts the main Jetpack Compose UI.
- AndroidManifest.xml (app/src/main/AndroidManifest.xml): Declares essential app configurations to the OS, including the package name, app icon, and the initial Activity to launch.
- App-level Gradle (app/build.gradle.kts): Configures module-specific build rules, such as SDK versions, enabling Compose, and declaring library dependencies.
- Version Catalog (gradle/libs.versions.toml): Centralizes dependency versioning, ensuring library updates are consistent across the entire project.
- strings.xml (app/src/main/res/values/strings.xml): Stores hardcoded text resources, keeping UI text separate from Kotlin logic to facilitate easier updates and localization.

Rotation Observation

I navigated to the second dimension, entered "Shuttle Tracker" into the text field, generated the feedback message, and then rotated the device.

Upon rotation, the UI completely reset. The screen returned to the first dimension, and the text field and feedback message cleared. All interactive state (index, input, feedback) was lost; only the default static UI survived.

This happens because Android destroys and recreates the Activity during configuration changes like rotation. Because my variables used standard remember rather than rememberSaveable, they only survived UI recompositions, not Activity recreation. Applications must preserve this state; otherwise, users lose unsubmitted forms and navigation progress every time they adjust their phone's orientation, causing deep frustration.

Mobile-Design Decision

To address mobile constraints, I displayed only one dimension at a time with "Previous" and "Next" navigation. A desktop application could easily display all six dimensions simultaneously using a grid. However, phones lack screen real estate. A single-item view prevents excessive scrolling, maintains legible font sizes, and keeps interactive touch targets (like buttons) comfortably within thumb reach, respecting the user's limited attention and mobile usage patterns.

Disclosures

Generative-AI Disclosure: I used Gemini in Android Studio. My prompts were "Solve the error and run the app" and "how should I run the app". Gemini suggested fixing a Kotlin 2.0 Compose Compiler issue by modifying `libs.versions.toml` and `build.gradle.kts`. I accepted the build script changes, verified them via a successful Gradle sync, and ran the app. I did not use AI to write my Kotlin UI logic.

